EduPro Course Demand & Revenue Forecasting

Executive Summary

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EduPro · Predictive Modeling for Course Demand and Revenue Forecasting

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Predictive Course Demand and Revenue Forecasting — EduPro

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Prepared for: Programme and funding stakeholders
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Scope: 60 courses · 10,000 transactions · 12 categories · 2025-01-01 to 2025-12-30 · total revenue 911,323

The finding

The data cannot support course demand forecasting, and we recommend not deploying one.

Enrolments are spread across the 60 courses exactly as they would be if the 10,000 purchases had been assigned to courses at random. Formally: the index of dispersion is 0.941 where 1.00 is pure chance, and a goodness-of-fit test returns p = 0.6046, far from the 0.05 that would indicate real differences between courses.

We tested 17 course characteristics — price, duration, rating, level, category, type and others — against enrolment. **None is associated with demand** once corrected for the number of tests run.

Six forecasting models were built anyway, to verify the diagnosis. The best scores **+0.009** on courses it had not seen, where 0.00 means "no better than guessing the average" and negative means worse.

The revenue number is arithmetic, not a forecast

Revenue forecasts in this dataset look excellent — R^2 up to +0.970. That figure is misleading and should not be reported as a modelling achievement.

Every transaction is paid at exactly the course's list price. There are no discounts or promotions in the data. So revenue is precisely *price* × *number of enrolments*, and since enrolments are near-constant, revenue is simply price × a constant.

We tested this directly. The rule **"revenue = list price × 166.7"** — no model, no machine learning, calculable in a spreadsheet column — predicts revenue with an average error of **1,724**. The best machine-learning model manages **3,017**, roughly 2 times worse.

If a revenue estimate is needed this quarter, the arithmetic rule is the better tool and it is auditable by anyone.

Why the data cannot answer the question

Three gaps, each fixable:

1. The catalogue does not record who teaches each course. The course list has no instructor column. The only link is on individual sales, where the median course shows 14 different instructors — that field identifies the transaction, not the teacher of record. Consequently the brief's instructor-quality analysis could not be performed at all.

2. Prices never vary. Each course has one price, always charged in full. Price sensitivity is measured from variation, and there is none to measure.

3. There is no measure of exposure. We can see enrolments but not how many people were shown each course. A course with few enrolments because it is unappealing and one with few because nobody saw it are indistinguishable here. This is likely the single most important missing field.

What we recommend

1. **Do not commission decisions from a demand model built on this data.** A forecast that is uncorrelated with outcomes but carries the authority of a model is worse than the honest retrospective reporting already in place.
2. **Use the arithmetic rule for revenue planning** in the interim, and state plainly that

it assumes flat demand across courses — which is what the data shows.

3. **Add course impressions or page views to the data collected.** Conversion rate (enrolments ÷ views) is the metric that would actually vary by course and would make this analysis viable.
4. **Add an instructor column to the course catalogue,** and record marketing spend and course launch dates against course IDs.
5. **Run one controlled price test.** Offering comparable courses at two price points is the only way to obtain a defensible price elasticity.
6. **Re-run this pipeline once those fields exist.** It is built, validated and reusable; the diagnostics that produced this verdict will equally confirm when real signal appears.

One caveat on this finding

"No signal detected" is not the same as "no signal exists". With 60 courses, a weak real effect could be present and undetectable. The conclusion is that **this dataset provides no evidence of predictable course-level demand**, not that demand is inherently unpredictable. The remedy is the same either way: measure more, then re-test.

The characteristics of the file — exactly five courses in every category, dispersion indistinguishable from random, and zero price deviation across 10,000 transactions — suggest it may be a generated dataset rather than a production extract. If so, the transferable output of this project is the method, which is complete and validated, rather than the specific finding.

Full methodology, diagnostics and model comparison: `research_paper.md`.
Reproducible via `python train.py && python generate_report.py`.